The Managerial Contribution of Coaches in the National Basketball Association *

Ramzy Al-Amine †

June 2020

Abstract

This paper explores the high frequency of front office changes to disentangle the relative contribution of head coaches in the National Basketball Association (NBA). By employing a manager-fixed effect regression approach, it estimates the average additional wins contributed by each head coach since 1985. Not only do the generated fixed effects provide much-needed quantitative insight into the relative effectiveness of current NBA head coaches, but they are also useful in predicting team wins for future coach-roster parings. I find that including coaching fixed effects reduces forecast errors by as much as 10 percent when predicting next season wins.

JEL-classification: C13, C23, D22, J44, J63

Key words: fixed effects, managerial contributions, organizational settings, sports, basketball.

*I thank Tim Willems, Kevin Wiseman, Dan Brown, Petr Sedlacek, Manasa Patnam, Tamim Bayoumi, Vladimir Klyuev and Kevin Mazur for useful comments.

†ramzyalamine@gmail.com

1 Introduction

Despite advancements in analytics, the basketball world continues to lack a systematic quantification of the coaching effect. Those with most silverware are traditionally considered all-time greats. But while championships are a signal for team success, the degree to which coaches contribute to them remains a matter of conjecture. For example, Phil Jackson's 11 championship rings are often over-looked in favor of perennial superstars (Michael Jordan, Kobe Bryant, Shaquille O'Neal). This paper accomplishes two objectives. First, it estimaties the relative contribution of coaches in light of roster characteristics and injuries. Second, it determines whether a coach's prior tenures are helpful in predicting his success in future tenures.

In line with the literature on managerial contributions in organizational settings, the empirical framework put forth in this paper relies on a coach-fixed effect methodology to control for roster characteristcs. The National Basketball Association (NBA) lends itself well to this methodology with the majority of coaches being observed on multiple teams throughout their careers. In addition, the availability of advanced perfromance metrics allow to controlling for roster quality. Results suggest that coaches can be as influential as All-Star caliber players. For example, changing from the mean head coach to Steve Kerr generates about 14 additional wins, on average, when roster characteristics are held at their means.

To validate the findings above, I examine whether the estimated fixed effects are useful in predicting future team wins. This is equivalent to asking the following question: for a coach who is observed on three teams throughout his career (e.g. Pat Riley), do the fixed effects estimated over two of the tenures (say Knicks, Lakers) predict success for the third (Heat)? I undertake a series of leave-one-out cross-validation excercises and find that models that include coach-fixed effects reduce forecast errors by as much as 15 percent relative to those that only inclue roster characteristics. These results are robust to different performance metrics and ways to capture roster ability.

This paper builds upon research across both sports and the corporate world. The literature on managerial contributions in organizational settings suggests that managers c

have come to the conclusion that coaches are inter-changeable (Gamson and Scotch; 1964, Cannella and Rowe; 1995, Fabianic; 1994, Smart and Wolfe; 2008, and Berri et al.; 2009). However, as shown in Berry and Fowler (2019), the potential coach effect may vary from one sport to another. For example, while the majority of studies on Baseball belong in the dismissive camp, Muehlheusser et al. (2016) find a wide degree of heterogeneous

effects among coaches in the German BundesLiga. These conclusions are consistent with those presented herein.

2 Data

The NBA is unique in its high frequency of front office changes, with around 60 percent of its coaches being observed on multiple teams—a much higher portion than in soccer, football, or baseball. The high turnover rate reduces standard errors as they move across teams. Moreover, the NBA provides advance metrics allow for capturing player ability levels (therefore controlling for roster quality).

The data set employed in this paper covers all regular seasons between 1985 and 2018. This comes down to 1,063 individual team-seasons and 165 head coaches. In addition to team wins, the variables considered for analysis include roster quality, games missed due to injury, and information on head coach identity. All the data is sourced from the Basketball Reference site. The average coaching tenure in the NBA extends to 2 seasons. Each coach is observed on 2.4 teams and 7.1 seasons, on average. Among notable coaches are Don Nelson and Lenny Wilkens whose careers span more than 30 seasons each, and Larry Brown who is observed on 11 different teams.

The structure of the NBA is such that no team possesses a systematic advantage over another. All teams are subject to a salary cap, forcing them to abide by a certain budget. [1] In addition, losing teams are compensated through the draft, ensuring (somewhat of) a balance of power. Although some big market teams may find a slight advantage in attracting free agents, it can be assumed that no franchise fixed effects are needed. The robustness section, however, considers further variations of the model.

2.1 Developing an indicator for roster quality

In order to control for roster quality, I construct a team-level indicator using advanced player metrics. The underlying assumption is that the best indicator for a player's ability level at the start of a given season is his overall performance during last season. As mentioned earlier, the basketball analytics world offers a variety of options for measuring overall performance. I choose Win Shares for ease of interpretability (although results are robust to the use of others—see Section 5.).

The roster quality indicator is calculated based on the following equation:

[1] The NBA adopts a *soft* salary cap, allowing teams to exceed it in some situations.

$$\text{RosterQuality}_{i,t} = \sum_{p=1}^{5} \text{WS}_{p,t-1} * \text{MP}_{p,i,t} \tag{1}$$

where $\text{WS}_{p,t-1}$ indicates player p's Win Shares in season $t-1$ and $\text{MP}_{p,i,t}$represents his total minutes played in season t under team i. [2] Weighing by total minutes played serves to account for players that are relied upon more heavily than others. Rosters are restricted to 5 players based on playing time.

This approach presents two main shortcomings. First, using previous year metrics as a proxy for coach-neutral ability level may exaggerate the effects of coaches joining young teams with budding superstars. Second, the formula may fail to account for highly-talented players either because they missed a significant part of last season or because they are in their first season in the NBA. I overcome this by ...

3 Analyzing the impact of head coach identity on team wins

The empirical framework consists in a coach-fixed effect (FE) regression model with team wins as a dependent variable. The variable of interest is a binary variable representing head coach identity [3]. As control variables, the model includes roster quality and injuries. Formally, this can be represented by the following equation:

$$\text{Wins}_{i,t} = \lambda_k + \beta_1 \text{RosterQuality}_{i,t} + \beta_2 \text{Injuries}_{i,t} + \epsilon_{i,t} \tag{2}$$

where $\text{RosterQuality}_{i,t}$ is team i's constructed roster ability indicator (using player performance metrics from season t-1), $\text{Injuries}_{i,t}$ is the total games missed by team i's top player (as indicated by the Win Share metric), λ_k is the coach identity variable, and $\epsilon_{i,t}$ is the error term. The coefficient on λ_k indicates the average additional wins contributed by coach k relative to the mean head coach. In the NBA, coaches do not usually make hiring decisions, therefore Ability can be assumed to be independent from λ_k. [4]

[2] I use overall win share from the season prior

[3] Equal to 1 for team i's head coach and 0 otherwise

[4] Except in very few cases where the coach was also assigned managerial powers (Doc Rivers—Clippers, Stan Van Gundy—Pistons, and Tom Thibodeau—Timberwolves).

3.1 Joint Impact of Coaches

Table 1 compares the results from adding coach-fixed effects onto a baseline regression model that includes the roster quality indicator and injuries as dependent variables. While both models point to high correlation between roster characteristics and team wins, the size of the coefficient on *RosterQuality* reduces by more than 35 percent when fixed effects are added. In contrast, the coefficient on *Injuries* only sees a 6 percent reduction in magnitude.

These results suggest that much of the players' impact on team wins is owed to the head coach. Indeed, by designing offensive and defensive systems, NBA head coaches are able to maximize on players' strengths and put their team in a position to win. A lot of the impact also happens beyond the court. Like managers in corporate settings, basketball coaches motivate and establish a culture. It is often through such intangible channels that they derive value especially from players with difficult personnalities or when from clashing superstars. For example, Phil Jackson and Pat Riley are known for their tremendous leadership abilities (just as much as their tactical brilliance). All of these channels translate into coaches being responsible of a sizeable portion of team wins.

Table 1: OLS regression (1), used to estimate Coach fixed effects

Dependent variable: Team Wins$_{i,t}$		
	Without Coach-FE	With Coach-FE
Roster Quality	$.58^{***}$ (9.46)	$.36^{***}$ (5.78)
Injuries	$-.31^{***}$ (-8.7)	$-.286^{***}$ (-4.46)
Constant	-40.18^{***} (-45.68)	-35.42^{***} (-42.51)
Coach fixed effects	No	Yes
adjusted R^2	.19	.39
F-statistic	97	4.9
N	814	814

Note: Estimates of OLS regression (1). Roster Quality is calculated as the mean roster win share from previous season. *Injuries* is calculated as the sum of games missed by the team's top 2 players by win share. Numbers in parentheses represent t-statistics, calculated using robust standard errors. * denotes significance at the 10% level, ** implies significance at the 5% level, *** indicates significance at the 1% level.

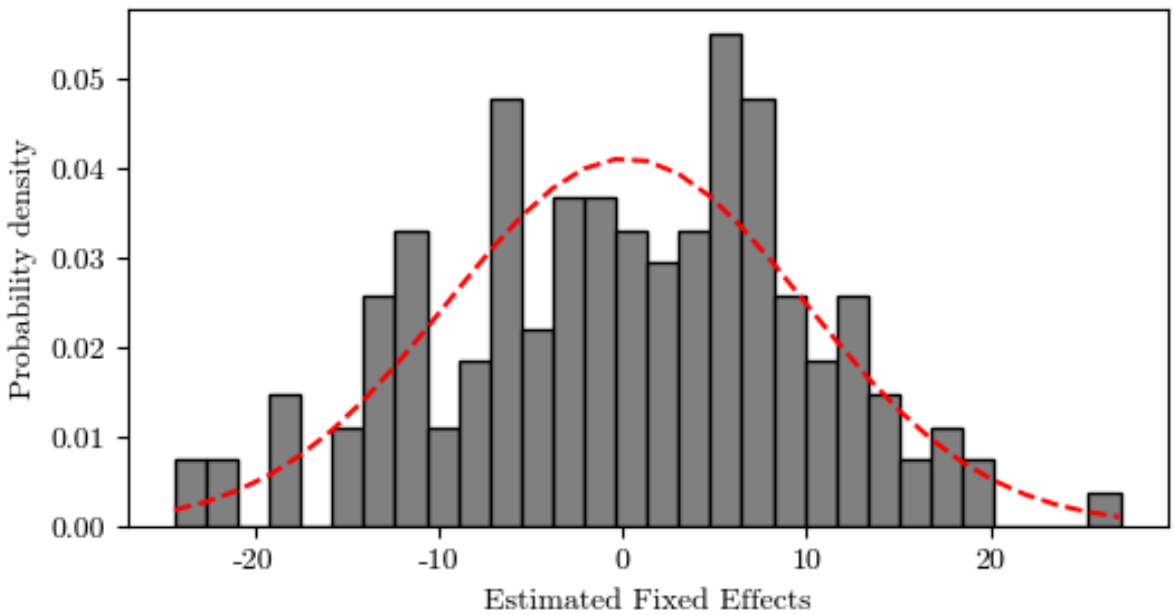

Figure 1: Probability density distribution of fixed effects

3.2 Individual Fixed Effects

Looking at the fixed effects for individual head coaches, the wide range of magnitudes suggests that the degree to which coaches impact the outcome of a basketball season differs considerably from one coach to another (Figure 1). The fixed effects are quite large for some coaches (as much as +20 wins per season)—enough to propel mediocre teams into playoff contention—and as low as -25 wins for others. Given the nature of fixed effects, these estimates represent deviations from a reference category, in this case taken as the mean head coach. For example, changing from the mean head coach to Steve Kerr generates around 12 additional wins, on average, when roster characteristics are held at their means (Figure 2). Meanwhile, changing to X generates around 4 additional wins.

The accuracy of these estimates is affected the number of seasons and teams under a coach's belt. Those with fewer observations are subject to higher standard errors. For instance, Bird, Ker, and Ainge, all of whom figure in the top 15th percentile of coaches, are limited to three observations each.[5] This means that the corresponding standard errors are high relative to coaches with more decorated careers (K.C. Jones, Gregg Popovich, Phil Jackson). This is illustrated by the size of confidence intervals in Figure 2. It is not suprising, however, that Bird, Ainge, and Kerr figure at the top ranking. Both Bird

[5]Bird and Ainge spent short tenures as head coaches before moving to General Manager positions. Kerr is a relatively recent head coach

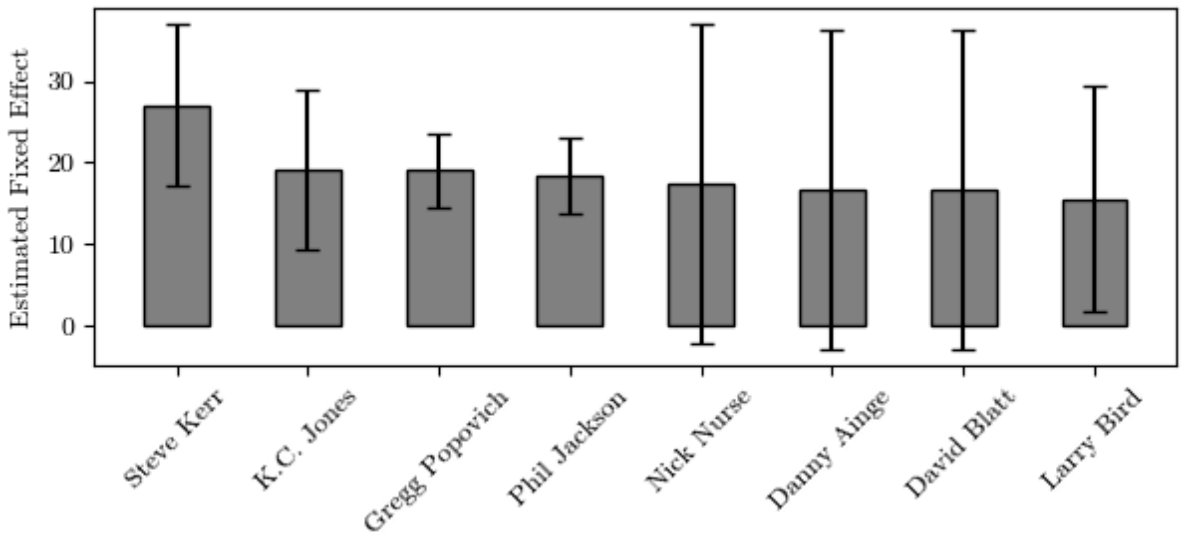

Figure 2: FE Point Estimate and Confidence Interval for Top 8 Coaches

and Ainge acquired exectutive positions (general manager) soon after the start of their coaching career, and Kerr reached the NBA finals in each of his first four coaching seasons.

To put the size of the estimates into perspective, it is helpful to compare against more player contributions to team wins. Overall, the impact of excellent coaches is likely to be within the sphere of that of All Star caliber players. For example, X's estimated fixed effect is of the size of allstar Damian Lillardâs win share metric in 2018-19 season. While the construction of this metric is different than fixed effects, the interpretation is similar. While this [6]. To put this in perspective, this amount of wins would have been enough to propel a team ranked 11th by conference standing during 2018-19 into finish sixth overall.

Table 2 compares the top 15th percentile against a *naive* approach of assessing coaching ability—win percentage (right-most two columns). Although some coaches figure at the top of both ranking approaches, many appear to be overlooked by standard win records. A case-in-point is Brad Stevens who is 53rd by win percentage and 8th in estimated fixed effect. Conversely, Mike Brown is 9th by win percentage and 42nd by estimated ability, which suggests that his success could have been aided by superior talent (LeBron James). This is testament that team performance alone does not accurately capture the coachâs contribution. While Stevens was able to elevate his roster to new heights, Brownâs case was aided by superior talent .

Table 2: Top 15th Percentile of NBA Head Coaches by Estimated Fixed Effect

[6]This comparison is for illustrative purposes only. The computational structure of the Win Share metric is different than the estimation methodology, yet interpretation is similar

Rk	Coach Name	Coeff.	t-Stat	Pct Rk	Win%	W-Pct Rk
0	Steve Kerr	27.04	5.41	1.00	0.79	1.00
1	K.C. Jones	19.14	3.84	0.99	0.68	0.97
2	Gregg Popovich	19.04	8.28	0.99	0.69	0.97
3	Phil Jackson	18.42	7.97	0.98	0.70	0.98
4	Nick Nurse	17.50	1.76	0.97	0.71	0.98
5	Danny Ainge	16.77	1.69	0.97	0.64	0.96
6	David Blatt	16.65	1.68	0.96	0.67	0.96
7	Larry Bird	15.56	2.20	0.96	0.74	0.99
8	Cotton Fitzsimmons	14.22	3.18	0.95	0.58	0.88
9	Tyronn Lue	13.67	1.38	0.94	0.61	0.92
10	Kevin McHale	13.45	2.69	0.94	0.55	0.84
11	Jerry Sloan	13.41	5.87	0.93	0.63	0.95
12	Mike Budenholzer	13.24	2.96	0.92	0.55	0.84
13	Billy Donovan	12.88	2.24	0.92	0.61	0.93
14	Scott Brooks	12.60	3.54	0.91	0.59	0.89
15	George Karl	12.22	5.34	0.91	0.59	0.91
16	Pat Riley	12.07	5.01	0.90	0.64	0.95
17	Erik Spoelstra	12.04	3.78	0.89	0.59	0.91
18	Del Harris	12.04	3.39	0.89	0.61	0.93
19	Tom Thibodeau	11.55	2.82	0.88	0.59	0.92
20	Doc Rivers	11.04	4.48	0.87	0.58	0.87
21	Rick Adelman	10.57	4.74	0.87	0.59	0.89
22	Chuck Daly	10.52	3.28	0.86	0.62	0.94
23	Mike Brown	10.43	2.73	0.86	0.62	0.94

Note: Estimates of Fixed Effect regression (1). $PctRk$ represents percentile ranking based on estimated fixed effect. $Win\%$ represents the coach's career winning percentage and $W - PctRk$ represents their percentile rank according to win percentage. Note: start year for career win percentage is 1985.

While these results point to the importance in idenitying the right manager, what makes a successful manager remains a matter of conjecture. Leadership experts emphasize that coaching is as much art as it is a science. However, there arestudies congrue that great coaches display a unique mix of character traits and a baseline skill level in the underlying activity. While character-based traits are difficult to assess systematically, I have looked into whether level of attainment as a player is a good predictor of coach's estimated fixed effect. Based on preliminary findings (see Appendix), there is no apparent

determinstic relationship there. In fact, the opposite might be true: those who were not previously skilled NBA players are more likely to succeed as NBA head coaches. While these findings may be interpreted as a result of relatibility, [7] more rigorous research is warranted as they are likely to be subject of survivorship bias. [8]

4 Predictive power of coach-fixed effects

I have established that some NBA head coaches are associated with considerably larger fixed effects than others, given roster characterisics and injuries. But for these generated estimates to have any real-world usefulness, they must have out-of-sample predictive power. This section aims to answer the following question: does a coach's estimated fixed effect over his career up to time t tell us anything about his team's success in time $t + 1$? The assumption is that, if these metrics have any out-of-sample power, then they could help inform hiring decisions for General Managers choosing from a pool of existing coaches.

The main procedure follows a standard leave-one-out cross-validation approoach, whereby the model is estimated on all observations *up to* time t, then applied to forecast next season wins (time $t + 1$) along with the roster quality indicator. I apply this procedure iteratively for each season from 1990 to 2019, computing error margins at each turn, then assessing the mean error across the sample as the primary evaulation crietria. For benchmark, results are compared against a standard model that does not include any FE (i.e. only the roster quality variable).

Results are summarized in Table 2. The model augmented with coach fixed effects (λ_k) reduces the RMSE by about 15 percent (from 9.12 to 8.53), on average, relative to a model with no fixed effects.

Figure 3 shows a boat.

Table 2: Predictive Power Evaluation Metrics

[7] Coaches who were secondary players are more likely to relate to their role players and derive the most value out of them

[8] This finding is likely subject to survivorship bias. From the current pool of NBA coaches, those were not previously excellent players are likely to have witnessed a more challenging path to become head coaches in the NBA, and have proven their abilities at the collegiate level or other leagues (or assistant roles) along the way.

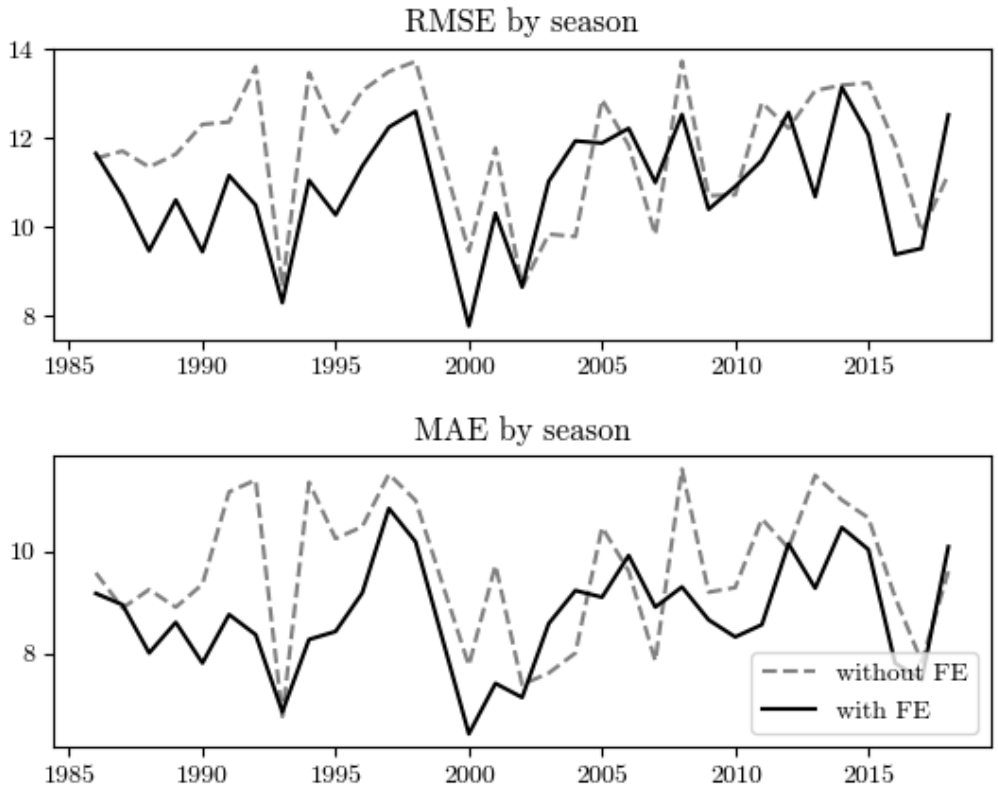

Figure 3: Forecast Errors

Dependent variable: Team Wins$_{i,t}$		
model	RMSE	MAE
Roster Quality	$.58^{***}$ (9.46)	$.36^{***}$ (5.78)
Injuries	$-.31^{***}$ (-8.7)	$-.286^{***}$ (-4.46)
Constant	-40.18^{***} (-45.68)	-35.42^{***} (-42.51)
Coach fixed effects	No	Yes
adjusted R^2	.19	.39
F-statistic	97	4.9
N	814	814

Note: RMSE generated based on LOOCV

5 Robustness

6 Conclusion

Reliance on past success records may overstate the true contributions of coaches who have had the fortune of inheriting superior players. Similarly, some coaches may fall short of expectations given roster characteristics. This paper addresses this problem by exploring the high frequency of front office changes in the NBA and advancing a robust empirical framework for disentangling the added-value of head coaches in light of roster characteristics and injuries. It finds that, just as acquiring a star athlete like LeBron James can boost team wins, hiring the right manager can elevate existing players to new heights.

To verify the accuracy and usefulness of the developed metrics, this paper shows that estimated fixed effects have predictive power. Specifically, models that include coach-fixed effects reduce forecasting errors by 10 percent on average, and as much as 15 percent. are useful in predicting team performance for newly formed coach-team pairs. In other words, In addition, it explores the effect of coachesâ backgrounds as ex-professional players and finds that, on average, little can be inferred about future coaching ability from a playerâs box score statistics.

These findings should be of value to general managers and executives in positions to make hiring decisions.

The paper adds to the line of research on managerial contribution by being the first to apply the fixed effect approach in the context of the NBA. This method is highly

utilized in scientific research, especially when analyzing the effect of manager identity on organizational performance. The basketball environment is particularly well-suited for this task since coaches are observed on multiple teams throughout their careers.

Some of the main shortcomings in this approach are summarized in the following. First, the relatively low number of observations for some coaches may over- or underestimate their average fixed effect. This issue would lessen as more historical data points become available for those who are relatively new. Second, influential factors other than coaching and player talent may be omitted, such as front office quality and training facilities. For the sake of the analysis, these factors are assumed to balance out across the league. Finally, as mentioned previously, the use of lagged performance metrics as a measure of roster quality may exaggerate the effects of coaches who join teams with young and budding superstars.